Alignment Unlocks Style: Distribution Arithmetic Reveals That Alignment Improves Rather Than Diminishes Stylistic Control

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Abstract

Alignment via instruction tuning is widely believed to reduce the diversity of language model outputs, imposing an "alignment tax" on stylistic and creative capabilities. We test this assumption directly by measuring stylistic impersonation ability across seven literary and functional styles in a paired base/aligned model setup (QWEN2.5-3B and QWEN2.5-3B-INSTRUCT). We then apply logit-space distribution arithmetic—linearly interpolating base and aligned model logits at each decoding step—to investigate whether base model distributions survive alignment and remain continuously navigable. Our results challenge the naive alignment tax narrative: the aligned model *significantly outperforms* the base model on style fidelity (7.26 vs. 3.74, $p < 0.0001$), and is the only condition that reliably differentiates between target styles (ANOVA $F = 6.32$, $p = 0.002$). Distribution arithmetic produces smooth, monotonic interpolation between base and aligned behavior, confirming that base distributions are neither destroyed nor discontinuously altered by alignment. We reframe the alignment–diversity tradeoff: base models possess diverse latent distributions but lack the controllability to access them on demand; alignment adds an addressing mechanism (instruction following) that unlocks this access. The relevant axis is not preservation versus loss, but controllability versus distributional freedom.

1 Introduction

Language models learn rich distributional structure during pretraining—including the ability to produce text spanning many registers, genres, and authorial voices. Alignment procedures such as RLHF Ouyang et al. [2022] and DPO Rafailov et al. [2023] reshape these distributions toward human preferences, yielding models that are helpful, harmless, and honest. But a persistent concern accompanies this process: *does alignment narrow the model's expressive range?* Kirk et al. [2023] showed that RLHF causes measurable mode collapse in both per-input and across-input diversity. Lin et al. [2023] documented degradation on downstream benchmarks, coining the term "alignment tax." If alignment suppresses stylistic distributions that existed in the base model, then creative applications—authorial impersonation, genre writing, code-switching across registers—may suffer.

But is this the right framing? The Superficial Alignment Hypothesis Zhou et al. [2024a] suggests that alignment is a thin veneer atop pretraining distributions, adjusting format and tone without fundamentally altering the model's knowledge or capabilities. Ji et al. [2024] provided further evidence: aligned models exhibit "elasticity," reverting toward pretraining distributions upon further fine-tuning. If alignment is superficial and reversible, then the base model's stylistic distributions should survive alignment intact—and distribution interpolation should be able to navigate continuously between base and aligned behaviors.

Our approach. We directly test these ideas by measuring stylistic impersonation across seven distinct literary and functional styles (Hemingway, Shakespeare, Poe, Austen, Tolkien, academic, noir) using paired base and aligned models (QWEN2.5-3B and QWEN2.5-3B-INSTRUCT). We apply logit-space distribution arithmetic— linearly interpolating base and aligned logits at each decoding step—at five interpolation coefficients ($\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$). A GPT-4.1 judge evaluates style fidelity and text quality, supplemented by automated diversity metrics (Distinct-N, Self-BLEU).

What we find. The results overturn the naive alignment tax hypothesis. The aligned model *significantly outperforms* the base model on style fidelity (7.26 vs. 3.74, $p < 0.0001$) and is the only condition that reliably differentiates between target styles (ANOVA $F = 6.32$, $p = 0.002$; base: $F = 0.87$, $p = 0.54$). Distribution arithmetic produces smooth, monotonic interpolation, confirming that base distributions survive alignment. But mixing in base logits *reduces* style quality rather than recovering it, because the base model lacks the instruction-following mechanism needed to address its own distributions.

In summary, our main contributions are:

- We conduct the first controlled comparison of stylistic impersonation ability before and after alignment, finding that alignment *improves* rather than diminishes controllable access to diverse styles.
- We apply logit-space distribution arithmetic to style generation, demonstrating smooth interpolation between base and aligned behavior across seven styles and three topics (231 total generations).
- We reframe the alignment–diversity tradeoff: the relevant axis is not preservation versus loss of distributions, but controllability versus distributional freedom. Alignment adds an addressing mechanism that unlocks latent stylistic diversity.

2 Related Work

Alignment and diversity loss. The observation that alignment reduces output diversity has been documented across multiple settings. Kirk et al. [2023] measured per-input and across-input mode collapse after RLHF on LLaMA-7B, finding substantial reductions in distinct n-gram diversity and Sentence-BERT similarity. Counterintuitively, increasing the KL penalty in PPO did not recover diversity—it reduced it further while also degrading performance. Lin et al. [2023] characterized the broader "alignment tax" on downstream benchmarks and found that simple weight-space interpolation between pre- and post-RLHF checkpoints ($\theta_{\text{merged}} = (1-\alpha) \cdot \theta_{\text{SFT}} + \alpha \cdot \theta_{\text{RLHF}}$) outperformed all other mitigation strategies, including KL penalties, LoRA, and experience replay. Our work differs from these studies in two ways: we focus specifically on *stylistic* diversity (targeted impersonation of distinct voices) rather than generic output diversity, and we operate in logit space rather than weight space.

KL-constrained alignment. DPO Rafailov et al. [2023] implicitly optimizes a KL-constrained reward objective with the closed-form solution $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \cdot \exp(r(x, y)/\beta)$, where β controls divergence from the reference policy. Korbak et al. [2022] showed that this framework is equivalent to Bayesian inference: the base model serves as the prior, the reward provides the likelihood, and the aligned model is the posterior. Larger β preserves more of the prior distribution. Extensions to alternative f-divergences Wang et al. [2023], Go et al. [2023] demonstrated that Jensen-Shannon divergence often provides better alignment–diversity tradeoffs than reverse KL. We build on this understanding by asking: regardless of which divergence was used during training, can we navigate between base and aligned distributions *post hoc*?

Distribution arithmetic and interpolation. Several lines of work have explored combining base and aligned model distributions at inference time. Zhou et al. [2024b] introduced emulated disalignment, showing that alignment can be arithmetically inverted via the log-linear combination $\pi_{\text{ED}}(y_t) \propto \pi_{\text{base}}(y_t)^{\alpha+1} / \pi_{\text{align}}(y_t)^{\alpha}$, effectively doubling the harmfulness of base models. Contrastive decoding Li et al. [2023] and proxy tuning Mitchell et al. [2023] similarly combine model distributions to steer generation. Most recently, Zhu et al. [2025] added a single identity-initialized layer trained with DPO, enabling continuous interpolation along a λ axis at inference time with only 2% of parameters trained. Lu et al. [2024] integrated model merging into every RLHF training step, showing that online weight averaging improves both preference scores and benchmark performance. Our contribution applies distribution arithmetic specifically to stylistic capabilities, testing whether these distributions are continuously navigable.

Superficial Alignment Hypothesis. Zhou et al. [2024a] proposed that alignment primarily teaches format and tone rather than new knowledge, suggesting that the base model's capabilities persist largely intact. Ji et al. [2024] provided mechanistic evidence: aligned models exhibit "elasticity," reverting toward the pretraining distribution upon further fine-tuning at a rate proportional to the ratio of pretraining to alignment data. Lake et al. [2024] further showed that alignment shifts diversity from distributional (diverse across samples) to Overton (diverse within a single response), and that base model behavior is recoverable via in-context learning. Our results provide direct evidence for the Superficial Alignment Hypothesis in the domain of stylistic generation: base distributions survive alignment, but alignment adds a crucial *addressing mechanism* for accessing them controllably.

3 Methodology

We compare three paradigms for accessing stylistic distributions in language models: (1) direct generation from the base model via completion-style prompting, (2) instruction-based generation from the aligned model, and (3) logit-space distribution arithmetic that interpolates between the two.

3.1 Models and Setup

We use QWEN2.5-3B as the base model and QWEN2.5-3B-INSTRUCT as the aligned model, both with 3 billion parameters. Each model is loaded in float16 on a separate NVIDIA RTX A6000 GPU (49 GB), enabling simultaneous inference for distribution arithmetic with KV caching.

3.2 Style Tasks

We select seven target styles spanning literary and functional registers: five authorial styles (Hemingway, Shakespeare, Poe, Austen, Tolkien) and two functional styles (academic writing, film noir narration). These styles were chosen to span a range of distinctiveness—from highly ornate and recognizable (Shakespeare, Poe) to understated and subtle (Hemingway)—allowing us to test whether some styles are easier to invoke than others.

Each style is applied to three topics: (1) "A person arriving at an old house for the first time," (2) "The experience of watching a sunset over the ocean," and (3) "A conversation between two strangers on a train." This yields $7 \times 3 = 21$ style–topic tasks.

3.3 Generation Conditions

For each style–topic task, we generate text under five conditions:

Base model ($\alpha = 0$). The base model receives a completion-style prompt describing the target style and topic. We generate 3 samples per task.

Aligned model ($\alpha = 1$). The aligned model receives a chat-formatted instruction requesting text in the target style. We generate 3 samples per task.

Distribution arithmetic ($\alpha \in \{0.25, 0.5, 0.75\}$). At each decoding step, we compute logits from both models on the same prefix and combine them linearly:

$$\ell_{\text{mix}} = (1 - \alpha) \cdot \ell_{\text{base}} + \alpha \cdot \ell_{\text{align}} \tag{1}$$

where ℓ_{base} and ℓ_{align} are the raw logit vectors from the base and aligned models, respectively, and α controls the mixture weight. Sampling then proceeds from $\text{softmax}(\ell_{\text{mix}})$. We generate 2 samples per task at each interpolation level.

All conditions use temperature $\tau = 0.8$, top-$p = 0.95$, maximum length of 128 tokens, and random seed 42. This yields $21 \times (3 + 3 + 2 \times 3) = 231$ total text samples.

3.4 Evaluation

LLM-as-judge. We use GPT-4.1 to evaluate each sample on two dimensions: *style fidelity* (1–10 scale: how well the output captures the target style's distinctive voice, vocabulary, and sentence

Condition	Style Fidelity	Text Quality	Distinct-2	Self-BLEU-4	Avg Len
Base ($\alpha = 0$)	3.74 ± 0.93	5.95 ± 0.65	0.882 ± 0.049	0.007 ± 0.012	103.2
$\alpha = 0.25$	5.14 ± 1.11	6.60 ± 0.73	0.913 ± 0.070	0.007 ± 0.016	104.4
$\alpha = 0.50$	6.36 ± 1.03	6.55 ± 1.22	0.884 ± 0.113	0.006 ± 0.016	99.8
$\alpha = 0.75$	6.67 ± 0.99	6.62 ± 1.35	0.915 ± 0.106	0.008 ± 0.020	97.1
Aligned ($\alpha = 1$)	$\mathbf{7.26 \pm 0.91}$	$\mathbf{7.07 \pm 1.16}$	$\mathbf{0.945 \pm 0.028}$	$\mathbf{0.003 \pm 0.004}$	91.4

Table 1: Aggregate results across all 7 styles and 3 topics. Style fidelity and text quality are GPT-4.1 scores (1–10). Distinct-2 measures lexical diversity (higher is better). Self-BLEU-4 measures intra-condition similarity (lower is better). Best results in **bold**.

structure) and *text quality* (1–10 scale: coherence and writing quality regardless of style). This required 210 API calls across all condition–task pairs.

Lexical diversity. We compute Distinct-N (type-token ratio for N-grams, $N \in \{1, 2, 3\}$) to measure vocabulary diversity within each sample. Higher values indicate more diverse word choices.

Intra-condition diversity. We compute Self-BLEU-4 (4-gram overlap between samples within each condition–style pair). Lower values indicate that samples within the same condition are more diverse from each other.

Style differentiation. We apply one-way ANOVA on style fidelity scores across the seven target styles within each condition. A significant F-statistic indicates that the model produces outputs that are distinguishable across styles—i.e., it is not generating the same generic text regardless of the style instruction.

Statistical tests. We use Mann-Whitney U tests for pairwise comparisons between conditions and report p-values with standard significance thresholds ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$).

4 Results

4.1 Main Results

table 1 presents the aggregate results across all styles and topics. The aligned model achieves the highest style fidelity (7.26 ± 0.91), significantly outperforming the base model (3.74 ± 0.93; Mann-Whitney U, $p < 0.0001$). Text quality follows a similar pattern, with the aligned model scoring 7.07 ± 1.16 versus 5.95 ± 0.65 for the base model. Lexical diversity (Distinct-2) is also highest for the aligned model (0.945), while Self-BLEU-4 is lowest (0.003), indicating that the aligned model produces both more stylistically varied and more lexically diverse text.

Smooth interpolation. Distribution arithmetic produces a monotonic increase in style fidelity from 3.74 (base) through 5.14, 6.36, 6.67, to 7.26 (aligned) as α increases. The curve is concave: the largest jump occurs between $\alpha = 0$ and $\alpha = 0.25$ (+1.40), with diminishing returns thereafter. At $\alpha = 0.75$, style fidelity (6.67) is not statistically distinguishable from the fully aligned model ($p = 0.051$). figure 1 illustrates this interpolation curve.

4.2 Style Differentiation

A model that reliably produces distinct styles should yield significantly different fidelity scores across target styles. table 2 reports one-way ANOVA results for each condition.

The base model produces undifferentiated outputs across all seven styles ($F = 0.87$, $p = 0.54$). Even with explicit style instructions in the prompt, the base model generates text that a GPT-4.1 judge cannot reliably distinguish across target styles. In contrast, the aligned model produces well-differentiated outputs ($F = 6.32$, $p = 0.002$). Intermediate interpolation levels show inconsistent differentiation, with marginal significance only at $\alpha = 0.25$.

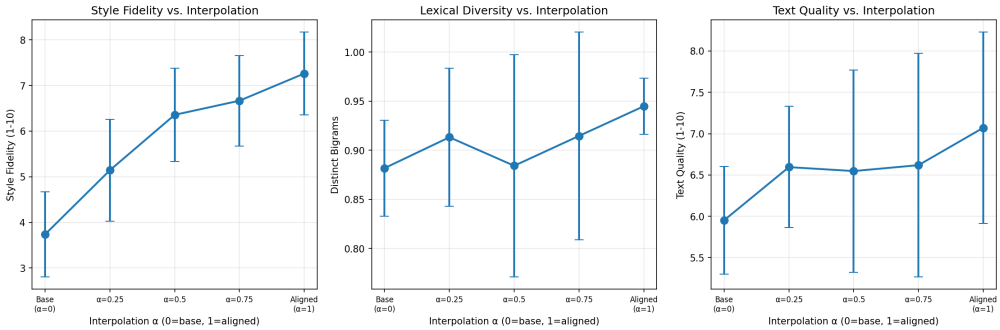

Figure 1: Style fidelity, text quality, and lexical diversity as a function of the interpolation coefficient α. Style fidelity increases monotonically with α, with a concave shape indicating that most gains come from the aligned model's instruction-following capability.

Condition	F**-statistic**	p**-value**	**Interpretation**
Base ($\alpha = 0$)	0.87	0.540	Not differentiated
$\alpha = 0.25$	3.03	0.041	Marginally differentiated
$\alpha = 0.50$	1.06	0.429	Not differentiated
$\alpha = 0.75$	2.12	0.116	Not differentiated
Aligned ($\alpha = 1$)	**6.32**	**0.002**	Well differentiated

Table 2: One-way ANOVA on style fidelity scores across 7 target styles within each condition. Only the fully aligned model reliably differentiates between styles ($p = 0.002$).

4.3 Per-Style Analysis

table 3 breaks down style fidelity by individual style. The aligned model outperforms the base model on every style, but the magnitude varies substantially. Ornate, distinctive styles—Poe ($+3.5$), noir ($+4.3$), Shakespeare ($+3.8$), academic ($+4.3$)—benefit most from alignment. The understated Hemingway style shows the smallest improvement ($+1.7$), suggesting that minimalist styles are harder to invoke even with instruction following.

4.4 Qualitative Examples

To illustrate the differences, we present representative outputs for two styles on the topic "A person arriving at an old house for the first time."

Hemingway style. The base model produced: *"The person arrives at the old house, a large one with a porch in the front and a garden in the back. The porch is weathered and creaks with every step."*—a generic description with no Hemingway character. The aligned model produced: *"Man walks up old cobbled stairs. Door unlocked. Pushes open. Inside dark. Candlestick near fireplace."*—capturing Hemingway's terse, declarative style. The $\alpha = 0.5$ condition yielded: *"She stood at the threshold, eyes wide. Doors creaked behind her, echoing softly. The walls were thick, heavy."*—showing intermediate characteristics.

Shakespeare style. The base model produced: *"I stumbled upon an ancient mansion, / Where the creaks and groans of the wooden floors / Echoed through the halls"*—modern poetry, not Shakespeare. The aligned model produced: *"In twilight's hallowed arms did he arrive, / At door of yore, where shadows doth entwine"*—with archaic vocabulary and iambic meter. At $\alpha = 0.5$: *"In hallowed halls, where echoes whisper past, / A stranger enters, stranger to the past"*—archaic vocabulary but modern rhythm.

Style	Base	$\alpha = 0.25$	$\alpha = 0.50$	$\alpha = 0.75$	Aligned	Δ (B$\rightarrow$A)
Hemingway	4.0	4.2	5.7	5.7	5.7	**+1.7**
Shakespeare	3.2	5.0	6.3	6.0	**7.0**	+3.8
Poe	4.5	6.5	7.0	7.7	**8.0**	+3.5
Austen	3.2	4.0	5.5	6.2	**7.0**	+3.8
Tolkien	4.2	5.5	7.2	7.3	**7.3**	+3.2
Academic	3.3	5.3	6.3	7.2	**7.7**	+4.3
Noir	3.8	5.5	6.5	6.5	**8.2**	+4.3

Table 3: Style fidelity by individual style across conditions. The Δ column shows the improvement from base to aligned. Ornate styles (Poe, noir, academic) benefit most from alignment; understated Hemingway shows the smallest gain.

5 Discussion

5.1 Reframing the Alignment Tax on Diversity

Our results challenge the common narrative that alignment diminishes stylistic capabilities. The base model has diverse latent distributions—this is well-established by the pretraining literature and confirmed by the smooth interpolation curves in our experiments. However, having a distribution and being able to *access* it on demand are fundamentally different capabilities. The base model scores 3.74 on style fidelity not because it lacks stylistic knowledge, but because it lacks the instruction-following mechanism needed to direct that knowledge toward a specific target.

This reframing aligns with the Superficial Alignment Hypothesis Zhou et al. [2024a] and the elasticity findings of Ji et al. [2024]. Alignment does not rewrite the model's distributional structure; it adds a thin control layer. The "alignment tax" on diversity documented by Kirk et al. [2023] is real at the level of *unsolicited* generation—aligned models default to a narrower range of outputs when not specifically instructed to diversify. But when *instructed* to produce a specific style, the aligned model draws on the same distributional knowledge as the base model, and does so far more effectively.

5.2 Distribution Arithmetic as a Diagnostic Tool

The smooth, monotonic interpolation curve (figure 1) serves as evidence that alignment operates continuously on the base distribution. There is no phase transition or discontinuity between base and aligned behavior—the two are connected by a navigable path in logit space. This is consistent with the log-linear structure exploited by Zhou et al. [2024b] and the continuous λ-interpolation of Zhu et al. [2025].

The concave shape of the curve is informative: most style improvement occurs between $\alpha = 0$ and $\alpha = 0.5$, suggesting that the aligned model's primary contribution is the instruction-following "address" rather than new stylistic knowledge. The aligned model does not know more about Hemingway than the base model does; it simply knows how to use what it knows when asked.

5.3 Why Some Styles Are Harder

The per-style analysis (table 3) reveals a pattern: ornate, additive styles (Poe, Shakespeare, noir) respond most to alignment, while the minimalist Hemingway style shows the smallest improvement. We hypothesize that this reflects an asymmetry in the generation process. Ornate styles require *adding* distinctive features—gothic imagery, archaic vocabulary, hard-boiled metaphors—which the instruction-following mechanism can reliably trigger. Hemingway's style, by contrast, requires *restraint*: shorter sentences, simpler vocabulary, the omission of elaboration. "Doing less" is a harder

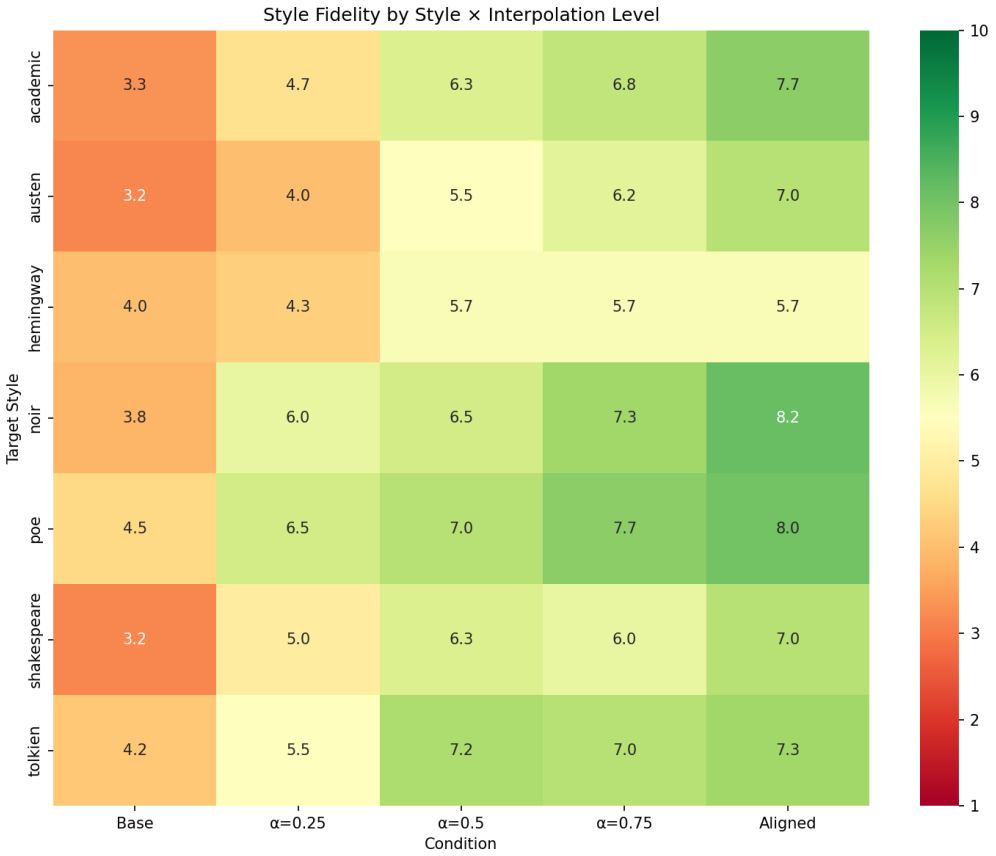

Figure 2: Style fidelity heatmap across styles and conditions. The aligned model (rightmost column) consistently achieves the highest fidelity, with ornate styles (Poe, noir) reaching scores above 8.

optimization target for autoregressive models, which are trained to predict the next token, not to suppress tokens.

5.4 Limitations

Single model pair. We tested only QWEN2.5-3B and QWEN2.5-3B-INSTRUCT. Results may differ for larger models, different model families, or models aligned with different methods (PPO vs. DPO vs. RLHF). Larger models may show different patterns due to greater distributional capacity.

LLM-as-judge bias. Style evaluation via GPT-4.1 is subjective and may carry systematic biases—for example, preferring fluent, well-structured text, which inherently advantages the aligned model. Human evaluation would provide a valuable complement, though it introduces its own biases regarding style recognition.

Benign styles only. Our seven target styles are all benign literary and functional styles. The more consequential case for the alignment tax hypothesis involves styles that alignment actively suppresses—toxic content, impersonation of real living persons, or other safety-relevant registers. Distribution arithmetic may show qualitatively different behavior in those settings.

Sample size. With 2–3 samples per condition–task pair, statistical power for per-style comparisons is limited. The aggregate results are robust ($N = 231$ total samples), but individual style estimates should be interpreted cautiously.

Prompt format confound. The base model received completion-style prompts while the aligned model used chat templates. This is a necessary design choice—each model performs best with its native format—but it means the aligned model's advantage may partly reflect superior prompt formatting rather than alignment *per se*.

Logit-space only. We tested only logit-space distribution arithmetic, not weight-space model averaging Lin et al. [2023]. Weight-space interpolation may exhibit different properties, particularly for layer-specific blending strategies.

6 Conclusion

We investigated whether alignment diminishes a language model's stylistic capabilities and whether logit-space distribution arithmetic can recover them. Our experiments on seven literary and functional styles, using paired base and aligned models (QWEN2.5-3B and QWEN2.5-3B-INSTRUCT), yielded a clear answer: alignment *improves* controllable access to stylistic distributions (style fidelity 7.26 vs. 3.74, $p < 0.0001$) rather than reducing it. The aligned model is the only condition that reliably differentiates between target styles (ANOVA $F = 6.32$, $p = 0.002$).

Distribution arithmetic confirms that base model distributions survive alignment intact: logit-space interpolation produces smooth, monotonic transitions between base and aligned behavior. However, the base model's contribution to the mixture is not "recovery" of lost capability but rather dilution of the control signal. The relevant tradeoff is not preservation versus loss, but controllability versus distributional freedom.

These findings suggest that future work on preserving diversity during alignment should focus less on preventing distributional loss (which does not appear to occur for stylistic capabilities) and more on maintaining or expanding the *addressability* of latent distributions. Promising directions include testing safety-suppressed styles where alignment may genuinely restrict access, comparing weight-space and logit-space interpolation, measuring unsolicited diversity (generation without style instructions), and investigating whether lightweight steering vectors can provide the addressing mechanism without full alignment.

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